Beepul Bharti

Data Science and AI Institute, Department of Biomedical Engineering, Johns Hopkins University
bbharti1@jh.edu | beepulbharti.github.io | beepulbharti

Research Interests

- Safe anytime-valid inference: e-processes, sequential testing, and online learning
- Uncertainty in learning: calibration, conformal prediction, and related methods
- Theory of responsible AI: interpretability, explainability, and algorithmic fairness
- Applications in high-stakes decision-making areas, including governance and healthcare

Education

2020– Ph.D. in Biomedical Engineering, Johns Hopkins University
Advisor: Dr. Jeremias Sulam
Relevant Coursework: Statistics, Matrix Analysis, Causal Inference, Nonlinear Optimization, Sparsity in ML/CV, Learning Theory, Replicable ML, Probabilistic Models in Vision

2016–20 B.A. in Mathematics & B.S. in Biomedical Engineering, Duke University
Honors: Cum Laude, Departmental Distinction
Relevant Coursework: Real Analysis, Abstract Algebra, Ordinary & Partial Differential Equations, Fluid Dynamics, Biostatistics, Probability, Multivariable Calculus

Experience

Sum' 2024 **Genentech**, Machine Learning Research Intern — San Francisco, CA
Deep Learning Theory and Algorithms Lab (DELTA)
- Explainability for motif discovery in genomic neural networks

Sum' 2017 **Duke University**, Bass Connections Fellow — Durham, NC
Department of Biostatistics
- Machine learning to predict schizophrenia-related hospital admission

Publications (*Indicates Equal Contribution)

2026 Parameter-Free and Group Conditional Online Conformal Prediction
Beepul Bharti, Ambar Pal, Jacopo Teneggi, Jeremias Sulam
Under review.

2026 Global Sequential Testing for Multi-Stream Auditing
Beepul Bharti, Ambar Pal, Jeremias Sulam
Under review.

2026 Let Me Explain, Again: Multiplicity in Local Sufficient Explanations
Ryan Pilgrim, Beepul Bharti, Jeremias Sulam, Rene Vidal
Transactions on Machine Learning Research (TMLR).

2025 Multiaccuracy and Multicalibration via Proxy Groups
Beepul Bharti, Mary Versa Clemens-Sewall, Paul Yi, Jeremias Sulam
International Conference on Machine Learning (ICML).

2025 Uncovering BioLOGICAL Motifs and Syntax via Sufficiency and Necessary Explanations
Beepul Bharti, Gabrielle Scalia, Alex Tseng
ICLR Workshop on Machine Learning for Genomics Explorations.

2024 Sufficient and Necessary Explanations (and What Lies in Between)
Beepul Bharti, Paul Yi, Jeremias Sulam
Conference on Parsimony and Learning (CPAL). **Oral**.

2024 Evaluation of AI Algorithmic Biases in Radiology
Paul H. Yi, Preetham Bachina, Beepul Bharti, Sean P. Garin, Adway Kanhere, Pranav Kulkarni, David Li, Vishwa S. Parekh, Samantha M. Santomartino, Linda Moy, Jeremias Sulam
Radiology.

2023 Estimating and Controlling for Equalized Odds via Sensitive Attribute Predictors
Beepul Bharti, Paul Yi, Jeremias Sulam
Advances in Neural Information Processing Systems (NeurIPS).

2023 SHAP-XRT: The Shapley Value Meets Conditional Independence Testing
Jacopo Teneggi*, Beepul Bharti*, Yaniv Romano, Jeremias Sulam
Transactions on Machine Learning Research (TMLR).

2023 Sex Imbalance Produces Biased Deep Learning Models for Knee Osteoarthritis Detection
David Li, Beepul Bharti, Jinchi Wei, Jeremias Sulam, Paul H. Yi
Canadian Association of Radiologists Journal.

Honors & Awards

2023 Alpha Eta Mu Beta: National Biomedical Engineering Honor Society
2022 JHU Mathematical Institute for Data Science Fellowship
2020 JHU Institute of Computational Medicine Fellowship
2016–2020 Duke University Dean's List
2019–2020 Duke University Pratt Fellowship
2019 Tau Beta Pi: The Engineering Honor Society

Teaching

Teaching Assistant

2023 EN.580.69: Biomedical Data Design, JHU
2021 EN.580.697: Computational Cardiology, JHU
2018 ECE 110L: Fundamentals of Electrical and Computer Engineering, Duke

Presentations

Talks

2026 Global Sequential Testing for Multi-Stream Auditing
UPenn CS Theory Seminar

2025 Sufficient and Necessary Explanations (and What Lies in Between)
Second Conference on Parsimony and Learning

Sufficient vs. Necessary Explanations
Machine Learning in Healthcare Club, University of New South Wales

2024 SHAP-XRT: The Shapley Value Meets Conditional Independence Testing
Explainable AI Seminars at Imperial College London

Algorithmic Fairness in Machine Learning and Data Science
EN.540.464: Advanced Biomedical Data Science for Biomedical Engineering

2023	Evaluating Fairness of AI Models in Radiology *Radiological Society of North America (RSNA) Annual Meeting*
	Fairness in Machine Learning *EN.540.464: Advanced Biomedical Data Science for Biomedical Engineering*
2022	Shapley Values and Hypothesis Testing *QMUL Intelligent Sensing Winter School*

Posters

2026	Global Sequential Testing for Multi-Stream Auditing *Safe Anytime-Valid Inference (SAVI) Conference*
2024	Certifying Fairness with Incomplete Sensitive Information *SIAM Conference on Mathematics of Data Science*
2023	Fairness via Sensitive Attribute Predictors *Columbia University Workshop on Fairness in Operations and AI*
	Fairness with Missing Sensitive Attributes *Johns Hopkins AI-X Foundry Fall Symposium*
	Shapley Values and Hypothesis Testing *Bern Interpretable AI Symposium*
2018	Using Machine Learning to Predict Schizophrenia-Related Hospital Admission *Duke School of Medicine Clinical Research Day*

Service

2025	Foreign Exchange Student Mentor *JHU Foreign Exchange Student Program*
2024	High School Student Mentor *JHU Whiting School of Internships in Science and Engineering (WISE)*
2023	Mentor Liaison *JHU Biomedical Engineering Application Assistance Program (BMEAAP)*

Reviewing

Reviewer for ICLR, ICML, TMLR, FAccT, ISIT, AISTATS, NeurIPS, and CPAL